

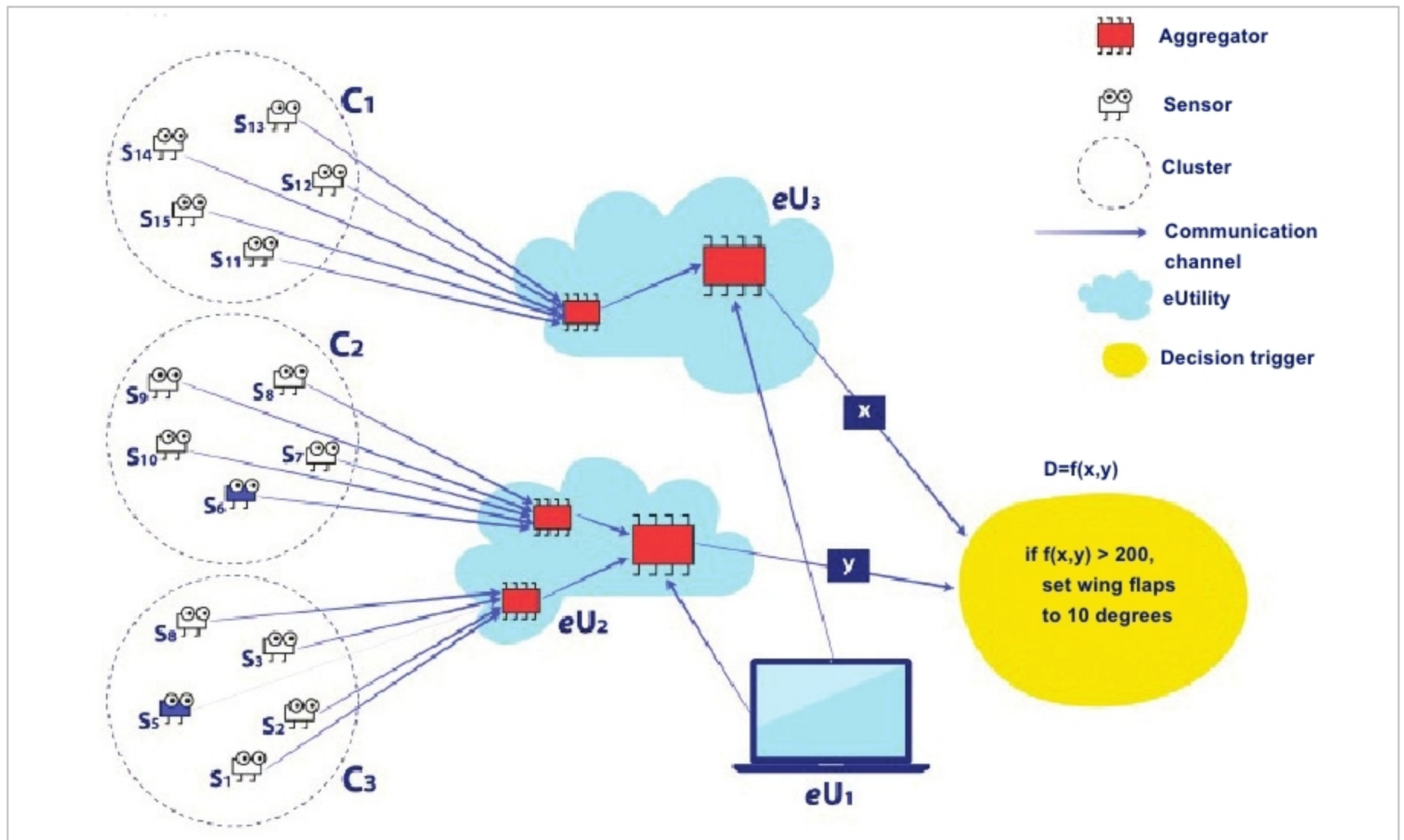


Figure 2: Constrained node network building blocks

Class	Network stack	Security
Class 0	No support.	No support.
Class 1	Specifically designed network stack for constrained nodes.	Supports security functions like authentication, confidentiality and integrity.
Class 2	Capable of supporting the same network stack as used on servers. Using the same network stack as used in Class 1 devices increases the interoperability.	Supports security functions like authentication, confidentiality and integrity.

Table 2: Support for the network stack and security

networks are deployed generally in the edge network of an IoT system.

The building blocks of a constrained node network are listed below.

- **Sensors:** Sensors are electronic devices, modules or sub-systems that measure physical properties and give an electrical output, e.g., temperature or acceleration sensors.
- **Actuators:** These are devices that take an electronic input and give a physical output, e.g., motors.
- **Clusters:** This is a grouping of sensors and actuators.
- **Communication channels:** This is a medium through which data is transferred, e.g., wired or wireless.
- **Aggregators:** These are the devices used to aggregate all the data from sensors and sometimes give commands to actuators. They are gateway devices.
- **eUtility:** This may be software, hardware or services that support aggregators in feeding data and in computing.

- **Decision trigger:** This is the software that does the computing and takes action if needed.

Special network stacks and constrained node networks

Constrained networks cannot use conventional TCP/IP stacks and need a new network stack because of their resource constrained nature. Refer Table 3 which compares protocols in conventional networks and constrained node networks.

Constrained node: Firmware comparison

The firmware for IoT devices need to meet the following requirements — energy efficiency, real-time capabilities, network connectivity, security, safety and a small footprint.

The firmware design relies heavily on which type of data link layer protocol is used in a constrained node. Table 4 gives a comparison between some of the commonly used link layer technologies.